

Seong-Jun Kang, Ph.D.

Experimental immunology, binding assays, and single-cell analysis
Postdoctoral Associate applicant · Keating Lab, MIT Department of Biology

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Available to start by December 2026

RESEARCH PROFILE

My research focuses on immune-cell states and function in inflammatory diseases, combining experimental immunology with computational analysis. My experimental work includes primary human T-cell functional assays, high-dimensional immune profiling, and single-cell and spatial transcriptomic experiments. I analyze gene expression and isoforms, integrate datasets, and evaluate predictions on held-out data using Python and R. In therapeutic development, I have evaluated antibody candidates, coordinated external studies, and contributed to non-human-primate preclinical studies and candidate-advancement decisions.

POSITION-RELEVANT EXPERTISE

- Binding measurements.** I ran Biacore SPR for association and dissociation kinetics, affinity ranking, receptor blockade, and human-primate cross-reactivity. I used ELISA binding, competition, and epitope assays alongside these measurements.
- Antibody candidate selection.** I compared antibody candidates in a humanized anti-CD40 program and followed lead selection with non-human-primate studies. The program supported a registered antibody patent in 2024.
- CRO coordination.** I designed and placed external studies, specified assay conditions, reviewed raw data, and tracked delivery dates. I used the results to make decisions about advancing candidates.
- Cellular and computational analysis.** I used high-parameter flow cytometry, single-cell and spatial profiling, and targeted proteomics to examine cellular responses. I wrote analysis code and tested results in held-out data.
- Grant writing and project execution.** I participated in proposal writing and project execution for industry projects totaling about \$1.3 million. I would consider LSRF and K99/R00 applications after confirming eligibility and project restrictions.

EDUCATION

DEGREES

Ph.D. 2014-2023	Seoul National University , Seoul, Republic of Korea Department of Biomedical Sciences <i>Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts</i> Advisor: Chung-Gyu Park, M.D., Ph.D.
M.S. 2012-2014	Seoul National University , Seoul, Republic of Korea Department of Biomedical Sciences <i>Thesis: Enhanced LPS-Induced Monocyte Activation by Recombinant Human Soluble CD14</i> Advisor: Chung-Gyu Park, M.D., Ph.D.
B.S. 2008-2012	Sungkyunkwan University , Suwon, Republic of Korea Department of Genetic Engineering

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher (postdoctoral) 2025-present	Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea Developed deep-learning approaches that combine imaging, transcriptomic, and clinical data in predictive models.
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Associate Director 2023–2025	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea I led the humanized anti-CD40 antibody program, including primate efficacy studies and regulatory documentation. I oversaw CRO and core-facility studies and made candidate-advancement decisions after reviewing data. I also built the in-house single-cell and spatial analysis pipeline.
Manager 2021–2023	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea I ran anti-CD40 antibody lead-selection research supported by two KDDF awards. I wrote the proposals for these awards, covering lead selection and optimization for multiple sclerosis and the PB101 efficacy and pharmacodynamic biomarker program for pemphigus vulgaris.
Research Student 2020–2023	Single Cell Atlas of Immune Diseases (SCAID) , Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea Contributed to the founding proposal and work within the funded program. Took part in milestone tracking, multi-site sample quality control, and coordination across principal investigators. The program covered more than 700 donors, eight organs, and 23 immune diseases.
Researcher 2019–2021	Xenotransplantation Research Center, Seoul National University , Seoul, Republic of Korea Processed rhesus macaque tissues and performed longitudinal immune monitoring in transplantation studies.
Teaching Assistant 2014–2016	Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea

PUBLICATIONS

18	17	1	6
TOTAL PAPERS	PEER-REVIEWED	IN REVISION	FIRST / CO-FIRST


[Google Scholar · Full Profile, Citations & h-index ↗](#)

PUBLICATIONS · FIRST / CO-FIRST AUTHOR

2026	Woogil Song*, Seong-Jun Kang*, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." <i>Nature Immunology</i>. in revision Co-first author. Isoform analysis of 251,055 memory CD4-positive T cells identified a PDE4D short-isoform bias in pathogenic Th1.17 cells. The study also tested selective inhibition.
2025	Seong-Jun Kang*, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8+ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." <i>Journal of Neuroimmunology</i>. Download PDF Co-first author. Immune-enriched single-cell analysis of 11,587 mouse brain cells examined CD8-positive resident memory T cells and their interactions with choroid plexus epithelium.
2025	Seong-Jun Kang*, Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." <i>Allergy</i>. Download PDF Co-first author. Single-cell profiling and functional assays examined skin-homing regulatory T cells in atopic dermatitis. OX40 was increased alongside impaired suppressive function.

2024	Jong-Min Kim*, Seong-Jun Kang* , So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." Xenotransplantation . Download PDF Co-first author. A tofacitinib-based regimen supported long-term diabetes control after islet allotransplantation in rhesus macaques. The work included longitudinal immune monitoring.
2022	Youngkyoung Lim*, Seong-Jun Kang* , Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." JEADV . Download PDF Co-first author. GeoMx profiling across 16 regions examined melanoma heterogeneity by invasion depth. HLA class I and II expression decreased with greater invasion.
2019	Jong-Min Kim*, Jun-Seop Shin*, Byoung-Hoon Min*, Seong-Jun Kang* , Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. "JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." Islets . Download PDF Co-first author. The study tested JAK3 inhibitor-based immunosuppression in cynomolgus macaques receiving allogeneic islet transplants, as an approach to controlling transplant rejection.

CONTRIBUTING AUTHOR

2026	Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, Seong-Jun Kang , Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype." Allergy . Single-cell RNA and TCR analysis identified a reactive, oligoclonal Th22 endotype in mycosis fungoides-like atopic dermatitis. The study distinguished this inflammatory state from malignancy.
2024	Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, Seong-Jun Kang , Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property." eBioMedicine . Download PDF High-dimensional immune profiling examined regulatory T cells in psoriasis. The study identified an impaired skin-trafficking phenotype in this population.
2024	Youngkyoung Lim*, Beom Keun Cho*, Seong-Jun Kang , Soyoung Jeong, Hyun Je Kim, Jiyeon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus." JEADV . Download PDF GeoMx spatial transcriptomics examined tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus, retaining the tissue locations of the measured gene expression.
2023	Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, Seong-Jun Kang , Chung-Gyu Park. "Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing RORyt through posttranslational modification." Experimental & Molecular Medicine . Mesenchymal stem cell-derived extracellular vesicles suppressed Th17 cells. The study linked this effect to destabilization of RORyt through post-translational modification.
2022	Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, Seong-Jun Kang , Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation." Experimental & Molecular Medicine . An optimized peptide inhibited calcineurin-mediated T-cell activation. The study examined peptide-based suppression of T-cell responses.
2021	So Hee Hong, Hyun Je Kim, Seong-Jun Kang , Chung-Gyu Park. "Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation." Current Diabetes Reports . Download PDF This review examined immunomodulatory approaches for porcine islet xenotransplantation, with a focus on strategies for controlling immune responses to transplanted islets.

- 2020** Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, **Seong-Jun Kang**, Chung-Gyu Park. "Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling." *Immunology Letters*.
Mesenchymal stem cell-derived exosomes suppressed T-cell proliferation. The study linked this effect to cell-cycle arrest through p27kip1/Cdk2 signaling.
- 2019** Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bongi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, **Seong-Jun Kang**, Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation." *Scientific Reports*.
Peripheral-blood RNA sequencing was used to investigate late graft loss after hyperglycemia in pig-to-primate islet transplantation, using bioinformatic analysis of blood gene expression.
- 2018** Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, **Seong-Jun Kang**, Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model." *Xenotransplantation*.
The study examined IL-6 blockade in a pig-to-primate islet transplantation model. Blocking IL-6 was associated with delayed revascularization of the transplanted islets.
- 2018** Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, **Seong-Jun Kang**, Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." *Xenotransplantation*.
Preclinical pig-to-primate islet transplantation was studied under an anti-CD40 antibody regimen using 2C10R4, examining this approach to immunosuppression in the transplant setting.
- 2018** Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, **Seong-Jun Kang**, Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." *Transplantation*.
The study evaluated a JAK inhibitor as a replacement for tacrolimus in cynomolgus macaques receiving allogeneic islet transplants.
- 2018** Jung-Sik Kim, Hyunwoo Chung, Nari Byun, **Seong-Jun Kang**, Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." *Biochemical and Biophysical Research Communications*.
The study constructed EMSC-islet composites for xenogeneic porcine islet transplantation. It examined a design that placed the two cell components together.

MANUSCRIPTS IN PREPARATION

- 2026** Kangsan Roh*, Sabyasachi Das*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." in preparation
Work in preparation examines ACTA2 base editing and lymphatic function in MSMDS. My contribution included Olink plasma proteomics and lymph-node immune-cell profiling.

■ TECHNICAL EXPERTISE

Binding kinetics and affinity

Hands-on **Biacore SPR** for association and dissociation kinetics, affinity ranking, receptor blockade, and human/non-human-primate cross-reactivity. **ELISA** binding, competition, and epitope analysis. Antibody candidate comparison and lead selection.

Therapeutic lead development	Humanized anti-CD40 antibody program from candidate comparison through non-human-primate studies. Assay design, external study coordination, raw-data review, regulatory documentation, and candidate-advancement decisions.
External studies and collaboration	CRO and core-facility study design, assay specifications, data review, and milestone tracking. Participation in outsourced antibody and AAV manufacturing collaborations involving Samsung Biologics and WuXi. Participation in early joint-development research with Merck, CSL, and Johnson & Johnson.
Cell, tissue & immunological assays	Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation. High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. Olink targeted plasma proteomics (immune & neurology panels).
Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; 10x 5'/3' library generation and 10x Chromium Connect automated preparation. Seurat , Scanpy , Squidpy , Monocle3 , CellChat/NicheNet ; scVI/scANVI, batch integration, consensus NMF ; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: GeoMx DSP run end to end, Visium , MACSima cyclic multiplexed immunofluorescence.
Computational	Python: scikit-learn, XGBoost/LightGBM/CatBoost, PyTorch ; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis.
Microscopy & animal models	Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.

BOOKS

Seong-Jun Kang, Do-gyun Kim, Jin-seon Han. **AI Drug Discovery: A Practical Guide for Medical and Biomedical Researchers** (title translated). Beommun Education. Forthcoming September 21, 2026. 548 pages. ISBN 979-11-5943-583-6. [Publisher](#)

The book covers molecular and protein data preparation, QSAR, ChemBERTa, graph neural networks, docking, virtual screening, candidate optimization, and follow-up experimental validation.

SELECTED GRANT WRITING & PROGRAM FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER – ASSOCIATE DIRECTOR)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Korea Drug Development Fund (KDDF)	Anti-CD40 antibody lead selection and optimization (multiple sclerosis)	2021–2023	~\$571,000	Lead proposal writer (>90% of proposal & execution)
Korea Drug Development Fund (KDDF)	Anti-CD40 lead compound (PB101): efficacy and pharmacodynamic biomarkers (pemphigus vulgaris)	2022–2024	~\$321,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022–2023	~\$222,000	Lead proposal writer; managed >90% of execution
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	~\$74,000	Lead proposal writer & core researcher

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
NAVER Corporation	Engineered regulatory T cells for immune-mediated diseases (PB20X_Treg)	2023–2026	~\$128,000	Proposal & study design (~50%)

DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of Science and ICT (MSIT)	Multi-layered immune-disease single-cell map for molecular target discovery (SCAID, A9-22-1-05)	2022–2028 (3+3)	~\$1.6M	Participated in proposal writing (>30%)
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory-Specialized Tech-Transfer & R&BD Grant	2020–2021	~\$40,000	Grant writing, experimental design & technical validation

I participated in proposal writing and project execution for the awards listed above. Budgets are approximate conversions at KRW 1,400 per USD. Project periods are award periods, not appointment dates.

AWARDS, PATENTS & SCHOLARSHIPS

PATENTS

- 2024** **Antagonistic Anti-CD40 Humanized Antibody.** Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor.
- 2019** **Immunosuppression Composition Comprising JAK Inhibitor.** Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20.

AWARDS

- 2024** **Grand Prize**, 2024 GPTers AI Hackathon (Project GAIA Team)
- 2023** **Grand Prize**, SBA Smart Workathon (Work Process Innovation)
- 2017** **Best Poster Award**, American Transplant Congress, Chicago · **Best Presentation Award**, Korean Association of Immunologists

SCHOLARSHIPS

- 2014–2020** NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship

SELECTED PRESENTATIONS

- Poster** KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012)

REFERENCES

Chung-Gyu Park, M.D., Ph.D.

Professor, Microbiology & Immunology / Biomedical Sciences, SNU College of Medicine; CEO/CSO, PB Immune Therapeutics · chgpark@snu.ac.kr

Doctoral advisor and industry supervisor; directed the anti-CD40 antibody program.

Hyun Je Kim, M.D., Ph.D.

Assistant Professor, Microbiology & Immunology / Biomedical Sciences; Interdisciplinary Program in Artificial Intelligence (IPAI), SNU · tte9801@snu.ac.kr

Co-corresponding on the single-cell studies; supervised the computational work.

Seung-Kuy Cha, Ph.D.

Professor, Dept. of Physiology & Global Medical Science,
Yonsei University Wonju College of Medicine ·
skcha@yonsei.ac.kr

Most recent research supervisor (2025–2026).

Jeong Seok Lee, M.D., Ph.D.

Associate Professor, Graduate School of Medical Science &
Engineering, KAIST, Daejeon · chemami@kaist.ac.kr

Corresponding author, PDE4D / Nature Immunology study.

Seong-Jun Kang · Curriculum
Vitae · Keating Lab

Available from August 2026 · based in Boston, MA · currently authorized to work in the United States
without employer sponsorship to begin employment

Updated
September 2026